1. What is Airflow?

Apache Airflow is a tool used to automate and manage tasks.

It helps us run different tasks automatically in the correct order without doing everything manually.

Example:

1. Extract data.
2. Clean the data.
3. Transform the data.
4. Load the data into a database.

2. Advantages of Using Airflow

1. **Automation:** Runs tasks automatically without manual work.
2. **Scheduling:** We can schedule tasks to run daily, weekly or hourly.
3. **Monitoring:** We can check whether a task is successful or failed.
4. **Error Handling:** If a task fails, Airflow can retry it.
5. **Time Saving:** Reduces manual work and saves time.
6. **Easy Management:** Helps manage multiple task in one place

3. What is DAG?

DAG stands for Directed Acyclic Graph.
A DAG is a workflow that shows which tasks should run and in what order.
Example:
Extract Data → Clean Data → Transform Data → Load Data

4. What is a Task?

A task is a single piece of work inside a DAG.
Each task performs one particular job.
Examples:

- Extract data from a file.
- Remove duplicate records.
- Clean missing values.

5. What is an Operator?

An operator tells Airflow what type of work a task should perform.

In simple words, an operator is used to create a task.

Common Types of Operators:

1. **PythonOperator:** Runs Python code.
2. **BashOperator:** Runs command-line commands.
3. **EmailOperator:** Sends emails.
4. **EmptyOperator:** Creates an empty task for workflow structure.

6. Difference Between DAG, Task and Operator

Name	Simple Meaning
DAG	Complete workflow
Task	One individual job
Operator	Defines what work the task performs